String s_1 = "shraddha"

String s_2 = "koli"

String name = s_1 + s_2

shraddha + koli

Shraddha koli

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9)

```
public class Main{
public static void main(String[] args) {
      String str1="NPTEL";
      String str2="java";
      int a=20;
      int b=10;
      System.out.println(str1+a+b); //Statement 1

      System.out.println(a+b+str2); //Statement 2
    }
}
```

Which of the following statement(s) is/are correct?

a. The output of the Statement 1 is NPTEL2010
b. The output of the Statement 1 is NPTEL30
c. The output of the Statement 2 is 1020java
d. The output of the Statement 2 is 30java

NPTL2010

x + y

↓

String → Concatⁿ

x + y + string

↓, Addⁿ

str1 + a + b

NPTL2010

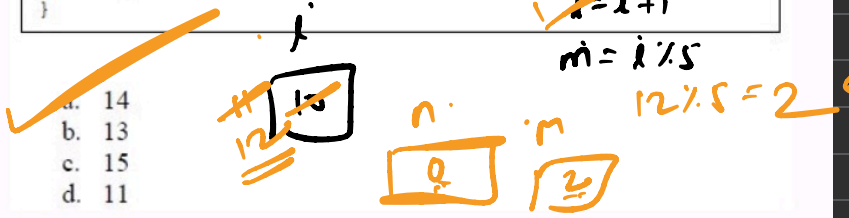
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1) **What is the output of the following program?**

```
public class Question
{
      public static void main(String[] args) {

         int i = 10;
         int n = i++%5;
         int m = ++i%5;
          System.out.println(i+n+m);
       }
}
```

a. 14
b. 13
c. 15
d. 11



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6) **Consider the following program.**

```
public class Question{
      public static void main(String args[]){
            for(int a=1;a<2;a+=1){
                  System.out.print(a+++a);
            }
      }
}
```

a + +a

a++ +a

What will be the output of the program if it is executed?

a. 3
b. 2
c. 4
d. 1

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CodeGreedy



Arrays in JAVA



CodeGreedy

“Success doesn’t come from what you do occasionally, it comes from what you do consistently.”

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WHY DO WE NEED ARRAYS

CodeGreedy

Imagine you want to store marks of 5 students:

```
int m1 = 90;
int m2 = 85;
int m3 = 78;
int m4 = 92;
int m5 = 88;
```

Managing 100 different variables becomes difficult.



x = 100, 1000

What if there are 1000 students?

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int $x_1 = 10$

int $x_2 = 20$

int $x_3 = 30$

$arr[] = \{ 10, 20, 30 \}$

$arr[] = \{ x_1, x_2, x_3 \}$

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WHAT IS ARRAY

CodeGreedy

Collection of same data type stored in contiguous memory

syntax

```
dataType[] arrayName = new dataType[size]
```

Example

```
int[] arr = new int[5];
```



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int arr[5]

} CPP / C

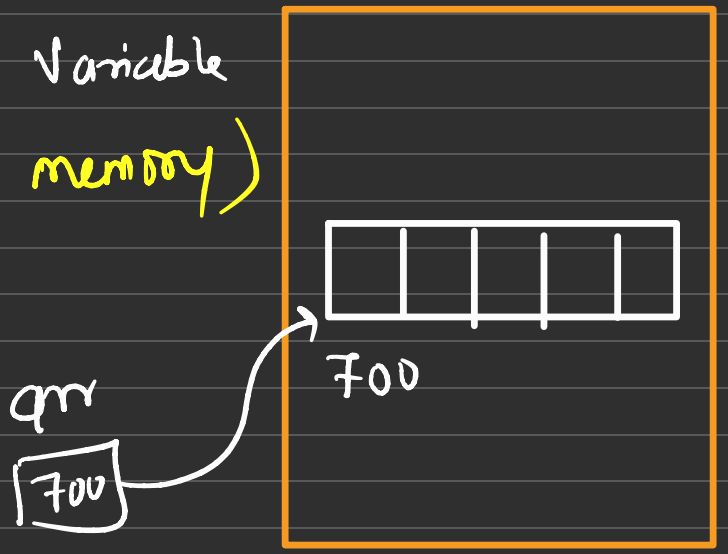


---

int []arr ⟶ Refrence Variable

(points to memory)

arr = new int[5]



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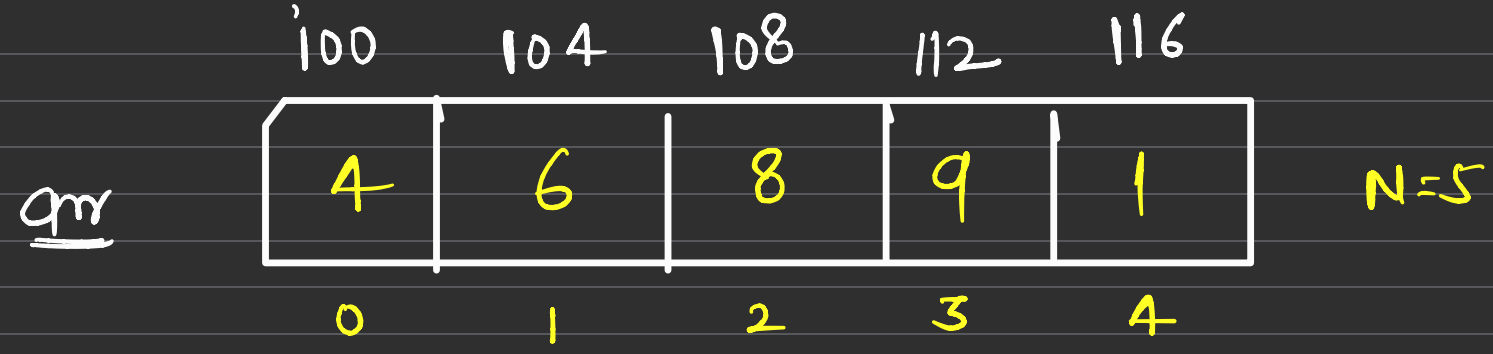
int [] arr; —— (Reference variable)

arr = new int[size]

1. int [] arr = new int[10]

2. int [] brr = { 10, 20, 30, 40 }

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arr(2)

arr + 2

100 + 2x4

100 + 8

108

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4B

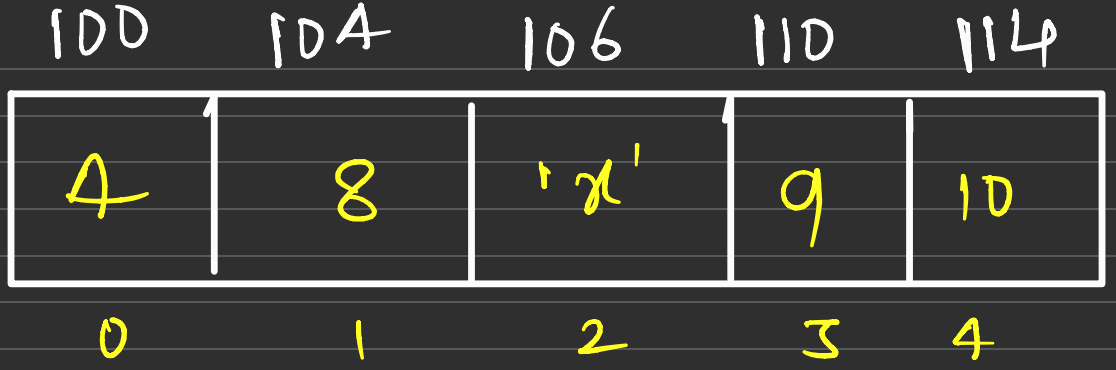


1B (C/CPP) ASCII (-128 to 127)
(0 to 255)

2B (Java) UNICODE ← 0 to 255 (ASCII)

16 bits 0 to 65535

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TYPES OF ARRAY

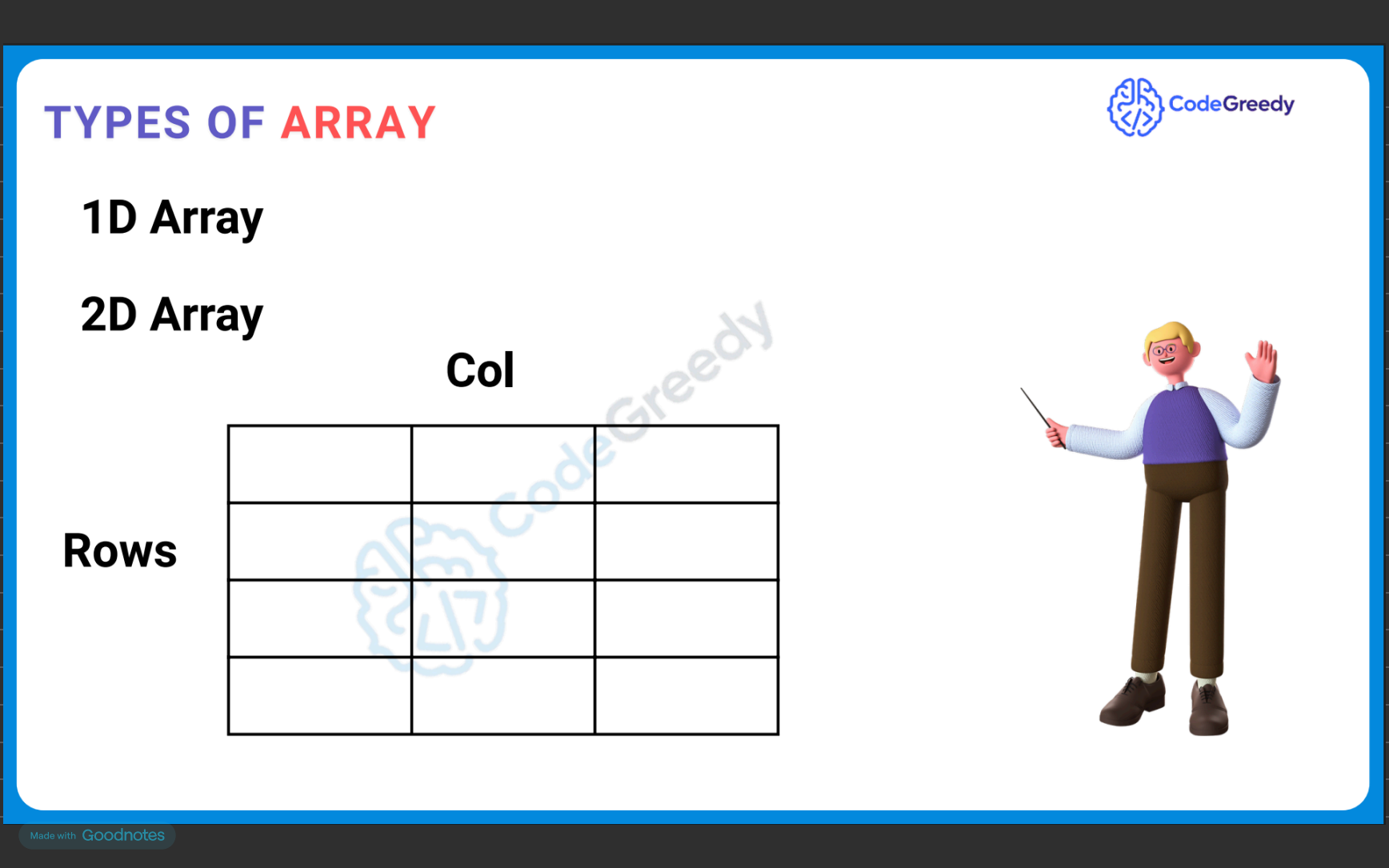
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1D Array

2D Array

Col

Rows



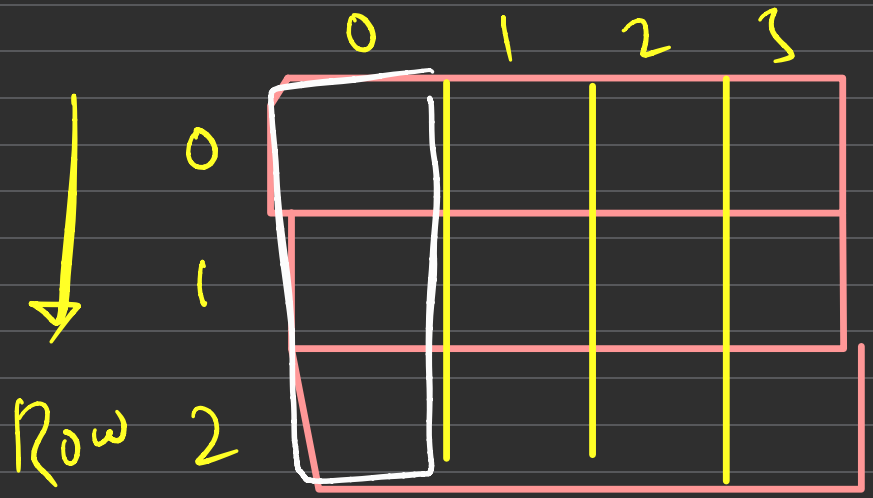
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```
int [] arr = new int [6]
int [][] mat = new int [3][4]
```

Row

↳ Column

⟶ Col

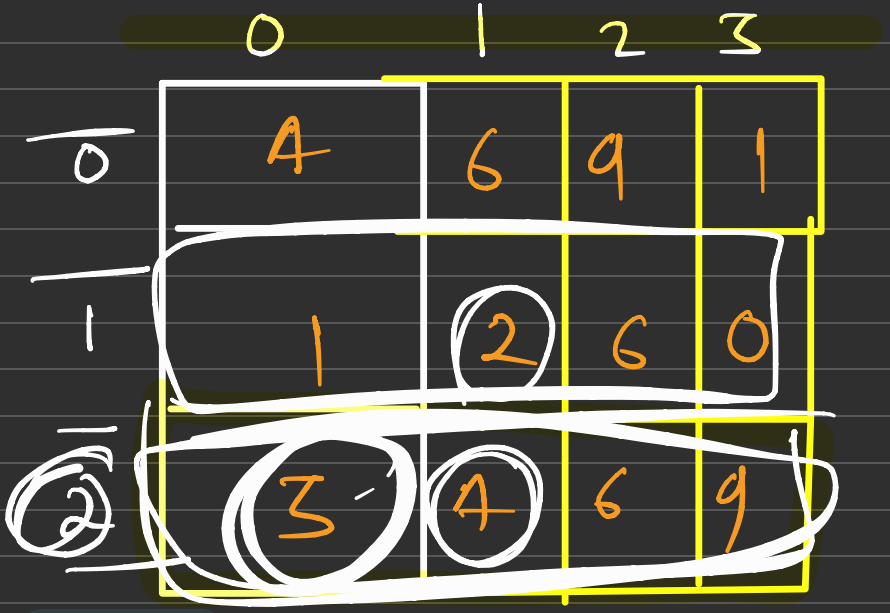


arr(3)

1

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3	1	0	6	2
0	1	2	3	4




```
for (i=0 ; i<3 ; i++)
{
    for (j=0 ; j<4 ; j++)
    {
        print (a[i][j])
    }
}
```

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INTERVIEW QUESTION

CodeGreedy

Why array always start from index 0



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TRAVERSING ARRAYS

CodeGreedy

Using for loop

```
for(int i = 0; i < arr.length; i++) {
   System.out.println(arr[i]);
}
```

for-each loop

```
for(int num : arr) {
   System.out.println(num);
}
```



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BASIC OPERATIONS ON ARRAYS

CodeGreedy

insert

```
arr[2] = 100;
```

Update

```
arr[1] = 50;
```

Earlier value get replaced by 50.



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COMMON INTERVIEW PROBLEMS

CodeGreedy

Find Maximum number from the array

Reverse the array

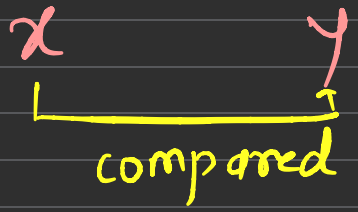
Check whether given number exist in the array.



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int x = 10, y = 20 (find max number)

```
if (x > y)
    print (x)
else
    print (y)
```



---

int x = 10, y = 20, z = 30 (find max number)



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N = 6

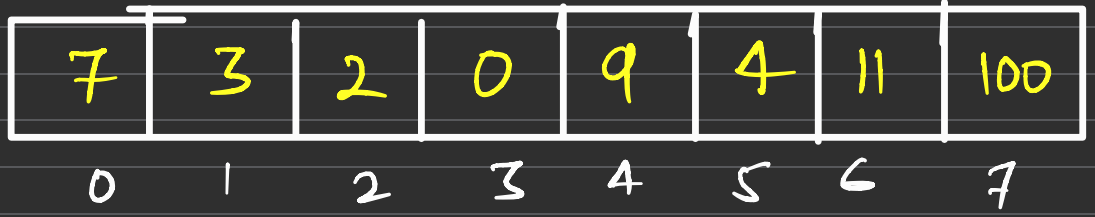
find largest No.

```
int max = a(0)
for (i=0 ; i<n ; i++)
{
    if (a(i) > max)
        max = a(i)
}
print (max)
```

find index of largest No.

```
int index = 0
int max = a(0)
for (i=0 ; i<n ; i++)
{
    if (a(i) > max)
    {
        max = a(i)
        index = i
    }
}
print (index)
```

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N=8

target = 11

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1. Two Sum

Solved

Easy | Topics | Companies | Hint | Re-do

You are given an array of integers `nums` and an integer `target`, return *indices of the two numbers such that they add up to* `target`.

You may assume that each input would have ***exactly* one solution**, and you may not use the *same* element twice.

You can return the answer in any order.

Example 1:

Input: `nums = [2,7,11,15], target = 9`

Output: `[0,1]`

Explanation:

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283. Move Zeroes

Solved

Easy | Topics | Companies | Hint | Re-do

Given an integer array `nums`, move all `0`'s to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

```
Input: nums = [0,1,0,3,12]
Output: [1,3,12,0,0]
```

Example 2:

```
Input: nums = [0]
Output: [0]
```

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121. Best Time to Buy and Sell Stock

Solved

Easy | Topics | Companies | Re-do

You are given an array `prices` where `prices[i]` is the price of a given stock on the ith day.

You want to maximize your profit by choosing a **single day** to buy one stock and choosing a **different day in the future** to sell that stock.

Return *the maximum profit you can achieve from this transaction*. If you cannot achieve any profit, return `0`.

Example 1:

```
Input: prices = [7,1,5,3,6,4]
Output: 5
Explanation: Buy on day 2 (price = 1) and sell on day 5
(price = 6), profit = 6-1 = 5.
Note that buying on day 2 and selling on day 1 is not
allowed because you must buy before you sell.
```

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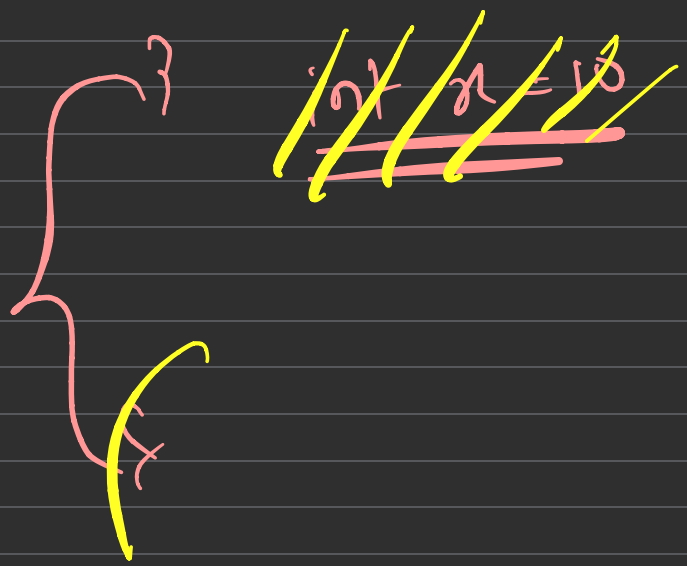
Storage classes

```
printf(Hello)
x = x++
printf(x)
```

Hello

Syntax for Var

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